

AERO50001 Aerodynamics 2 Problem Set #1

Extra practice: Isentropic relations and compressible flow

1.1 Mach number

☆☆☆ 3-5 mins

An aircraft is flying at 320 m/s in the standard atmosphere ($\gamma = 1.4$, $R = 287 \text{ J Kg}^{-1} \text{ K}^{-1}$).

<i>Geometric Altitude (km)</i>	<i>Pressure (N/m²)</i>	<i>Temperature (K)</i>	<i>Density (kg/m³)</i>
0	1.0133 E + 05	288.150	1.2250 E + 00
1	8.9875 E + 04	281.651	1.1117 E + 00
2	7.9501 E + 04	275.154	1.0066 E + 00
3	7.0121 E + 04	268.659	9.0926 E - 01
4	6.1669 E + 04	262.166	8.1934 E - 01
5	5.4048 E + 04	255.676	7.3643 E - 01
6	4.7217 E + 04	249.187	6.6012 E - 01
7	4.1105 E + 04	242.700	5.9002 E - 01
8	3.5651 E + 04	236.215	5.2578 E - 01
9	3.0800 E + 04	229.733	4.6707 E - 01
10	2.6500 E + 04	223.252	4.1351 E - 01

- (a) Calculate the Mach number M if the aircraft is flying at an altitude of 2 km.
(b) Calculate the Mach number M if the aircraft is flying at an altitude of 10 km.
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$$(a) M_2 = \frac{u}{a_2} = \frac{u}{\sqrt{\gamma R T_2}} = 0.962$$

$$(b) M_{10} = \frac{u}{a_{10}} = \frac{u}{\sqrt{\gamma R T_{10}}} = 1.068$$

1.2 Isentropic relations

★☆☆ 3-5 mins

Air ($\gamma = 1.4$) is flowing isentropically in a channel. At one point, the Mach number is 1.3. The stagnation pressure and temperature are $P_0 = 500$ kPa and $T_0 = 350$ K.

- (a) Calculate the pressure and temperature using the compressible flow tables.
 - (b) Calculate the pressure and temperature using the isentropic relation equations.
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(a) When $M = 1.3$

$$P = P_0 \left(\frac{P_0}{P} \right)^{-1} = \frac{P_0}{2.7707} = 180.46 \text{ kPa}$$

$$T = T_0 \left(\frac{T_0}{T} \right)^{-1} = \frac{T_0}{1.3380} = 261.58 \text{ K}$$

$$(b) P = P_0 \left(1 + \frac{\gamma-1}{2} M^2 \right)^{\frac{-\gamma}{\gamma-1}} = 180.45 \text{ kPa}$$

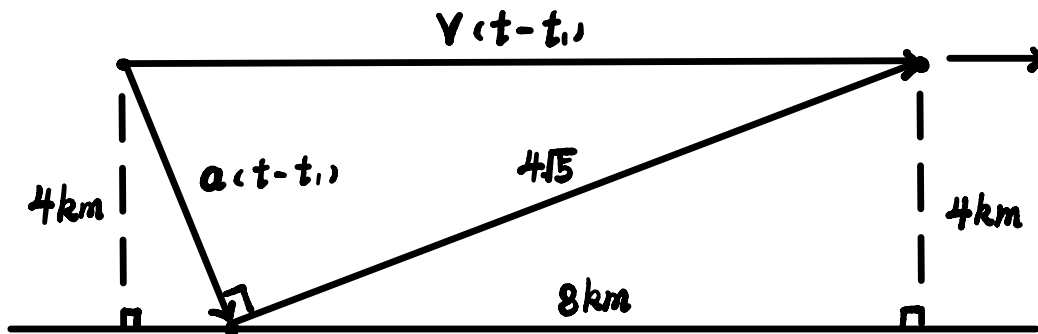
$$T = T_0 \left(1 + \frac{\gamma-1}{2} M^2 \right)^{-1} = 261.58 \text{ K}$$

1.3 Mach number and wave angles

☆☆☆ 5-8 mins

A person standing on the ground does not hear a supersonic aircraft flying at an altitude of 4000 m until it is 8 km past them.

What is the aircraft speed in m/s?



$$\text{Let } V(t-t_i) = v, \quad a(t-t_i) = \alpha$$

$$\therefore v^2 = \alpha^2 + (8^2 + 4^2) = \alpha^2 + 80$$

$$(v-8)^2 + 4^2 = \alpha^2 = v^2 - 16v + 80$$

$$\therefore v^2 - 80 = v^2 - 16v + 80$$

$$\therefore v = 10 \text{ (km/s)}$$

$$\therefore \alpha = 2\sqrt{5} \text{ (km/s)}$$

$$\text{At } 4\text{km}, (t-t_i) = \frac{\alpha}{a} = \frac{\alpha}{\sqrt{8RT}} = 13.76 \text{ (s)}$$

$$\therefore v = \frac{v}{(t-t_i)} = 726.72 \text{ (m/s)}$$

1.4 Isentropic relations at the throat

★☆☆ 10-15 mins

A perfect gas is expanded isentropically from a reservoir where the pressure is p_0 , the density is ρ_0 and the absolute temperature is T_0 . At the throat $M = 1$.

[Note: for exponentials, use ** instead of ^.]

- (a) Obtain an expression for the pressure ratio $\frac{p}{p_0}$ at the throat.
 - (b) Obtain an expression for the density ratio $\frac{\rho}{\rho_0}$ at the throat.
 - (c) Obtain an expression for the temperature ratio $\frac{T}{T_0}$ at the throat.
 - (d) Calculate these ratios for air ($\gamma = 1.4$)
 - (e) Calculate these ratios for Helium ($\gamma = 1.66$)
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$$(a) \frac{p}{p_0} = \left(1 + \frac{\gamma-1}{2}\right)^{-\frac{\gamma}{\gamma-1}}$$

$$(b) \frac{\rho}{\rho_0} = \left(1 + \frac{\gamma-1}{2}\right)^{-\frac{1}{\gamma-1}}$$

$$(c) \frac{T}{T_0} = \left(1 + \frac{\gamma-1}{2}\right)^{-1}$$

$$(d) \text{ for air, } \frac{p}{p_0} = 0.528$$

$$\frac{\rho}{\rho_0} = 0.634$$

$$\frac{T}{T_0} = 0.833$$

$$(e) \text{ for Helium, } \frac{p}{p_0} = 0.488$$

$$\frac{\rho}{\rho_0} = 0.649$$

$$\frac{T}{T_0} = 0.752$$